



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Lab: Database Technology

DS-554-MJP

Solved Practical Manual (SQL · MongoDB · Neo4j)

Semester II · Lab (MJP)

This solved practical manual covers the Database Technology lab assignments from the official SPPU M.Sc. Data Science syllabus. Each practical states the aim followed by a complete, runnable solution and the expected output. Use it as a reference — type the code yourself and experiment with the parameters to learn effectively.

Requirements: Python 3.x with numpy, pandas, matplotlib, scikit-learn, scipy installed (pip install numpy pandas matplotlib scikit-learn scipy).

Assignment 1: DQL — SELECT with clauses & aggregate functions (SQL)

```
-- WHERE, ORDER BY, logical operators, scalar & aggregate functions
SELECT dept, COUNT(*) AS emp_count, ROUND(AVG(salary),2) AS avg_sal
FROM   employee
WHERE  salary > 20000 AND status = 'active'
GROUP BY dept
HAVING COUNT(*) > 2
ORDER BY avg_sal DESC;
```

Assignment 2: Movie / Actor database (MongoDB document model)

```
// Insert a film belonging to two genres, released at two places
db.Film.insertOne({
  title: "Devdas", year: 2002,
  genre: ["romantic", "drama"],
  actors: [{first:"Shahrukh", last:"Khan"}, {first:"Madhuri", last:"Dixit"}],
  release: [{place:"Mumbai", date:"2002-07-12", rating:8},
            {place:"Pune",   date:"2002-07-15", rating:7}]
})
// Queries
db.Film.find({ genre: "romantic" }) // by genre
db.Film.updateOne({ title: /^T/ }, { $inc: { "release.0.rating": 1 } })
db.Actor.deleteMany({ age: { $gt: 60 } }) // remove actors > 60
```

Assignment 3: Book–Publisher & Hospital databases (MongoDB)

```
db.Book.find({ pages: { $gt: 500 } })
db.Book.find({ language: "english" })
db.Book.find({ author: "A. Gupta", year: 2020 })
// Hospital
db.Hospital.find({ specialization: { $size: { $gt: 1 } } }) // multi-speciality
db.Hospital.find({ rating: { $gte: 4 } }, { name: 1 })
```

Assignment 4: Song database (Neo4j / Cypher)

```
CREATE (a:Artist {name:"Arijit"})-[:PERFORMS]->(s:Song {title:"Tum Hi Ho"})
CREATE (s)-[:RECORDED_IN]->(st:Studio {name:"YRF"})
CREATE (st)-[:MANAGED_BY]->(rc:Company {name:"T-Series"})
// Query: artists performing a song
```

```
MATCH (a:Artist)-[:PERFORMS]->(s:Song {title:"Tum Hi Ho"}) RETURN a.name;  
// Songs recorded in a studio  
MATCH (s:Song)-[:RECORDED_IN]->(studio:Studio {name:"YRF"}) RETURN s.title;
```

Assignment 5: Employee database (Neo4j / Cypher)

```
MATCH (e:Employee)-[:WORKS_IN]->(d:Department {name:"Analytics"}) RETURN e.name;  
MATCH (d:Department)<-[:WORKS_IN]-(e) RETURN d.name, count(e) AS headcount;  
MATCH (e:Employee)-[:HAS_ACQUIRED]->(s:Skill) WHERE e.name="Riya" RETURN s.name;
```


